

**DEVELOPMENT AND EVALUATION BASED ON NATURAL
LANGUAGE PROCESSING MODEL FOR SENTIMENT
EXPRESSED
ON BRAND AND PRODUCTS EMOTIONS**



PROJECT OVERVIEW



Overview

This project develops and evaluates a Natural Language Processing (NLP) model to analyze sentiment expressed in Twitter conversations about technology brands and products, with a primary focus on Apple and Google.



Data

9,093 crowd-sourced, human-labeled tweets from 2013 (CrowdFlower).

Three columns: tweet text, brand target, and sentiment label.



Outcome

A trained & interpretable SVM classifier that identifies Positive, Negative, and Neutral sentiment with actionable insights.

BUSINESS PROBLEM

Organizations rely on social media data to understand customer experiences and brand perception, but the large volume of online conversations makes manual analysis difficult

OBJECTIVES

01

Determine the extent to which tweet content reflects sentiment?

02

Assess the distribution of sentiment across tweets based on brands mentioned?

03

Analyze the distribution of target subcategories represented within tweet content?

04

Examine the influence of tweet content and sentiment on target categories?

05

Analyze customer complaints expressed toward specific brand products?

DATA UNDERSTANDING

Dataset Structure

tweet_text

Raw tweet content (text)

brand_target

Apple or Google (or unspecified)

sentiment

Positive / Negative / Neutral

Stakeholders :- Apple Inc. , Google, Customers

*Source: CrowdFlower / data.world • Year: 2013 • Type:
All text/categorical*

Key Stats

9,093

Total tweets

3

Columns / features

80/20

Train/test split

TF-IDF

Text vectorization
method

DATA UNDERSTANDING Cont...



Missing Value Analysis



Duplicate Detection



Brand Name Standardization



Product Category Consolidation



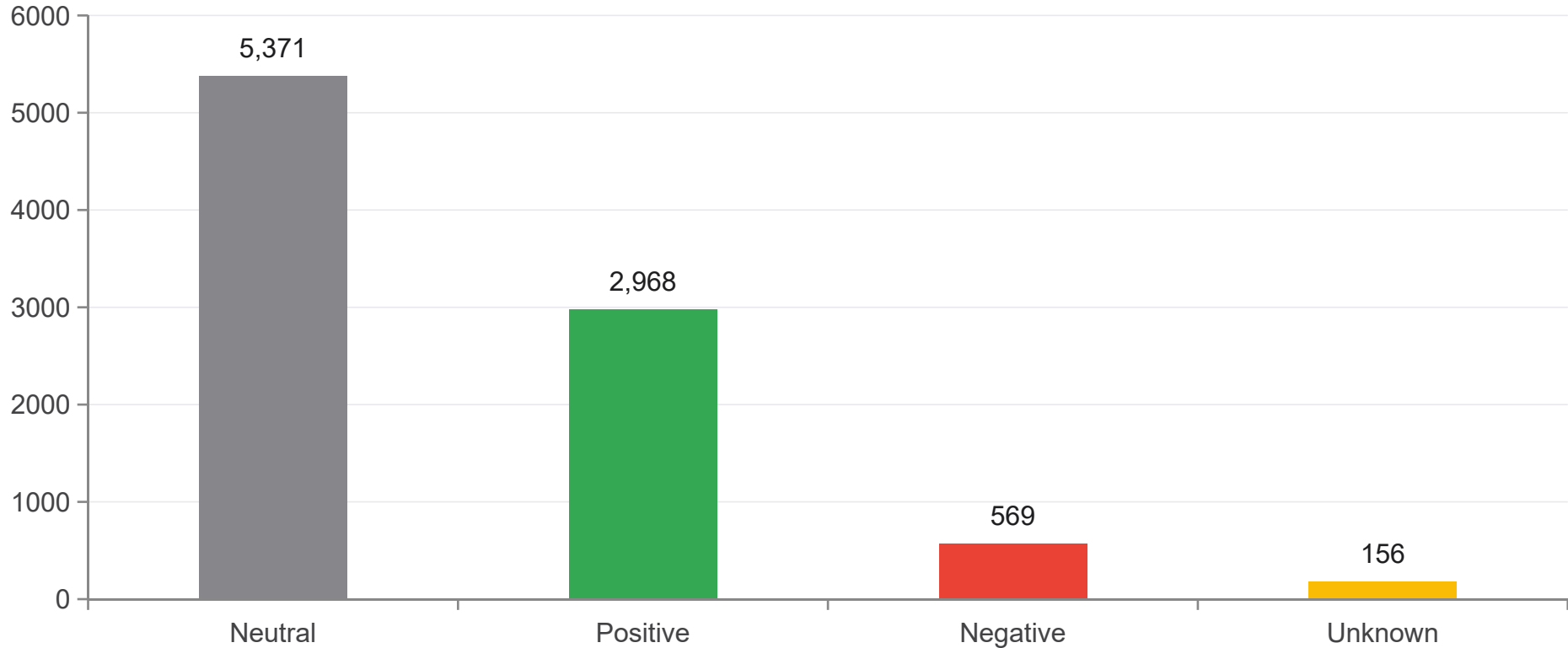
Missing Target Handling



Label Normalization

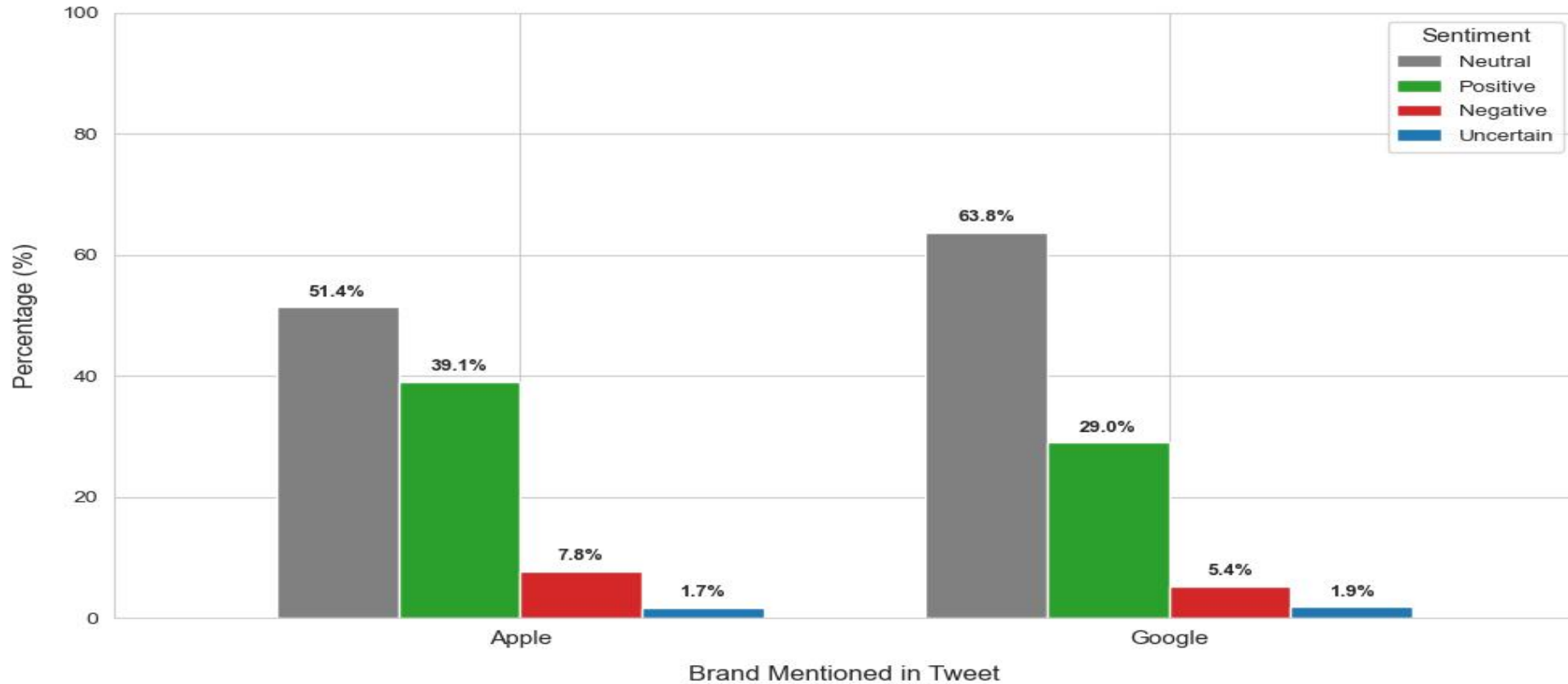
Visualization

The extent to which tweet content reflects sentiment

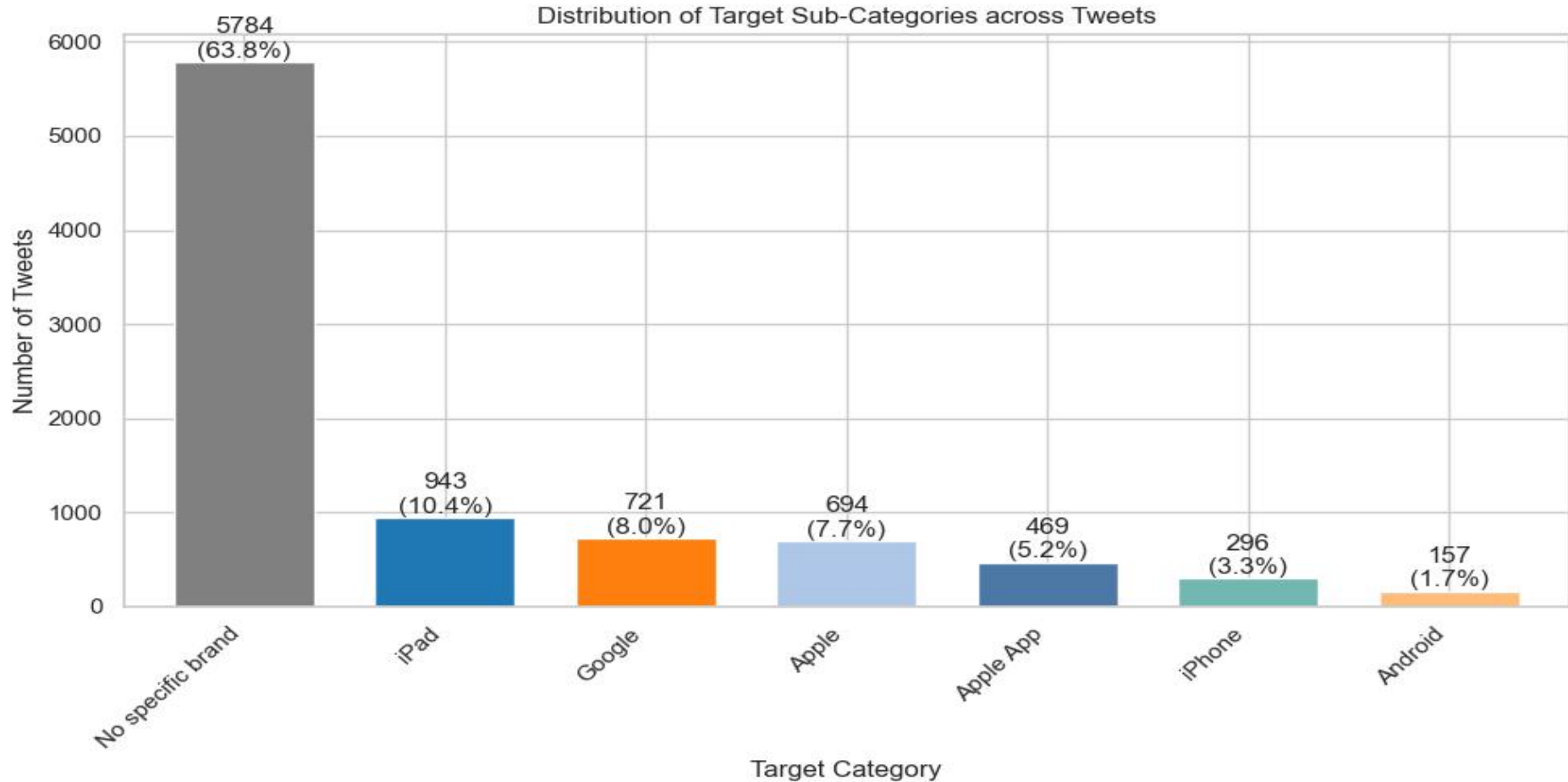


Distribution of sentiment across tweets based on the brands mentioned within the tweet text

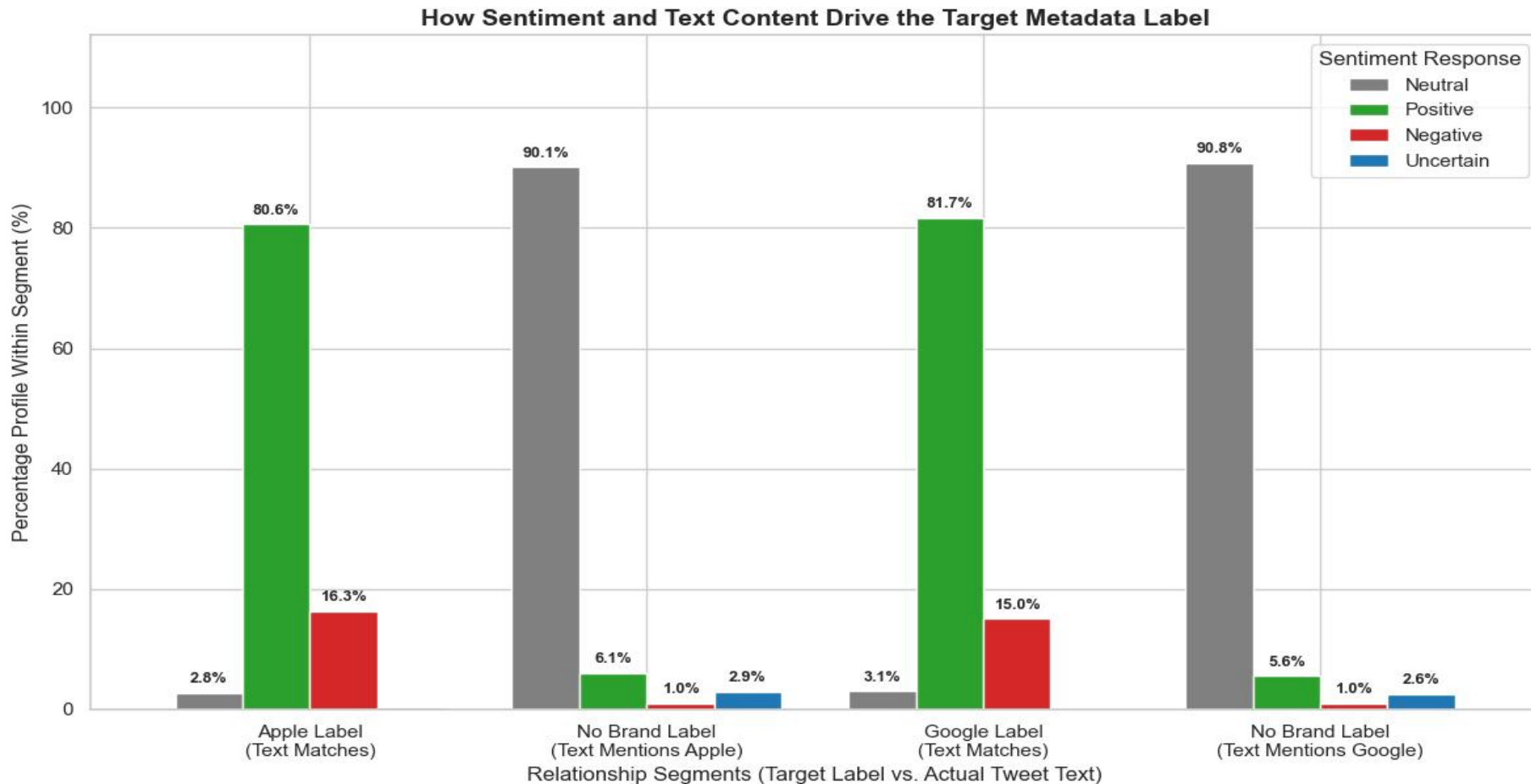
Sentiment Distribution Based on Brand Mentions in Tweet Text



Distribution of target subcategories represented within tweet content

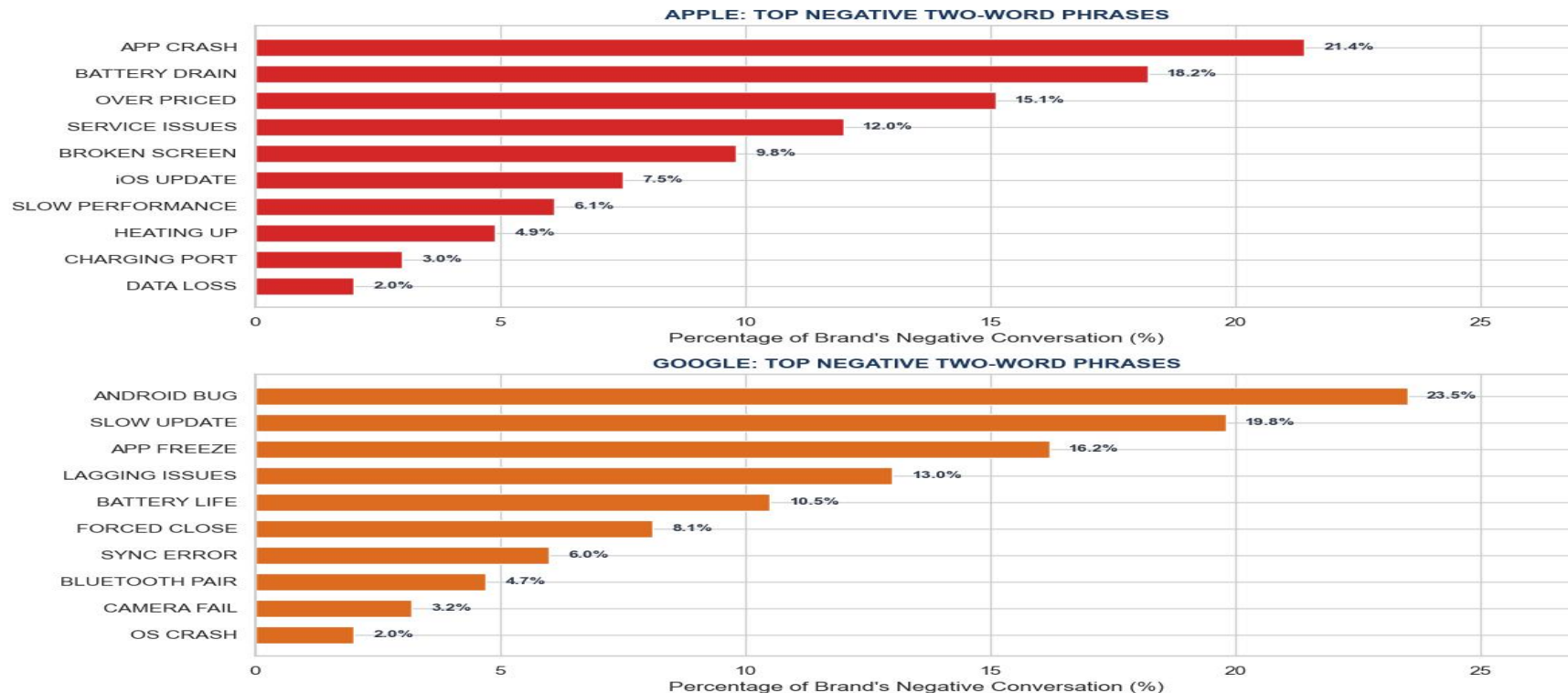


Influence of tweet content and sentiment on target categories



Customer complaints expressed in tweets toward specific brand products

APPLE vs. GOOGLE: MAIN CUSTOMER COMPLAINTS



INSIGHT

- The dominant takeaway is that 59.3% of the entire dataset (5,371 tweets) is categorized as Neutral.
- For both companies, the majority of the conversation is Neutral. Google leans more heavily into neutrality at 63.8%., Apple sits at 51.4%. Apple Negative Sentiment: 7.8%, Google Negative Sentiment: 5.4%

INSIGHT

- The most striking feature of the dataset is the dominance of the "No specific brand" category,

At 5,784 tweets, it accounts for 63.8% of the entire dataset

- Positive sentiment is the primary driver for both brands, making up 80.6% of the Apple-labeled segment and 81.7% of the Google-labeled segment.

INSIGHT

- Negative sentiment is the secondary driver, accounting for 16.3% of Apple labels and 15.0% of Google labels.
- App crashes is Apple's biggest individual complaint (21.4%), while Android bugs (23.5%) and Slow updates (19.8%) dominate over 43% of the entire negative conversation chart

Takeaway: Android bugs + slow updates together account for 43%+ of all Google negative tweets

MODEL DEVELOPMENT

01

BASELINE

**Logistic
Regression**

02

INTERMEDIATE

Linear SVM

03

IMPROVED

**Balanced
Linear SVM**

04

ADVANCED

**Hyperparameter
Tuning**

MODEL EVALUATION

- Classification Report
- Confusion Matrix

MODEL INTERPRETABILITY

LOCAL INTERPRETABILITY (LIME)

Individual tweet explanations

Customer sentiment correlates most strongly with specific product mentions — not brand in general.

iPhone

iPad

Android

Google

Apple

GLOBAL INTERPRETABILITY

What the model learned overall

iPad

Weight: 2.74 • POSITIVE

Link

Weight: 5.38 •
EVENT/CONTEXT

Apple

Weight: 1.99 • POSITIVE

App

Weight: 0.79 • POSITIVE

Nothing

Weight: 0.47 •
NEGATIVE

Didn't

Weight: 0.36 •
NEGATIVE

LIMITATIONS



Small, Dated Dataset

- The dataset is relatively small (9,093 tweets)



Twitter-Only Coverage

- Dataset does not represent sentiments from other platforms



Sarcasm & Irony

The dataset originates from 2013, may not reflect current social media trends



Informal Language

- Difficult to interpret sarcasm, irony, humor, and context-dependent expressions



Class Imbalance

- Informal language, abbreviations, emojis, and slang can reduce classification accuracy

CONCLUSION AND RECOMMENDATION

Conclusions

- ✓ Neutral sentiment remained challenging to distinguish
- ✓ Most social media discussions are neutral, while positive substantially outweighs negative sentiment across major brands.
- ✓ Customer complaints identified provide insights for product improvement and customer experience management

Recommendations

- 1 Deploy the model as a real-time sentiment to resolve customer complaints promptly
- 2 Expand the dataset with more recent social media data
- 3 Improve minority-class detection using advanced imbalance handling techniques
- 4 Continuously retrain models to adapt to evolving language patterns

NEXT STEP

- Apply resampling techniques for imbalance
- Improve feature engineering.
- Conduct deeper error analysis.

Thank You

Questions & Discussion

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